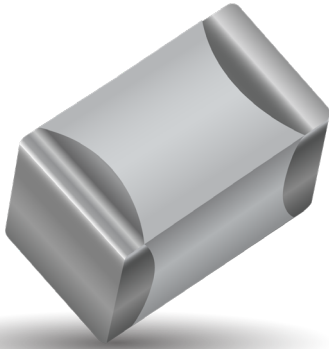


RF/Microwave Capacitors

RF/Microwave Multilayer Capacitors (MLC)

800B Series NP0 Porcelain, High RF Power Ultra-Low ESR



GENERAL DESCRIPTION

KYOCERA AVX's 800B Series offers superb performance in demanding high RF power applications requiring consistent and reliable operation. The combination of highly conductive metal electrode systems, optimized case geometries, and proprietary dielectrics, yields the lowest ESR. KYOCERA AVX's new NP0 low loss rugged dielectrics are designed to provide superior heat transfer in high RF power applications. Ultra-low ESR and superior thermal performance insure that the 800B Series products are your best choice for high RF power applications from VHF through microwave frequencies.

TYPICAL APPLICATIONS

- Avionics
- Public Safety Radio
- Wireless Communications
- VHF / UHF / HDTV Broadcast Transmitters
- Telecom
- WiMAX
- Microwave Communication Systems and Satellite Systems

TYPICAL CIRCUIT APPLICATIONS

- High RF Power Filter Networks
- Matching Networks
- Couplers
- Combiners
- Output Coupling
- Antenna Coupling
- DC Blocking
- Bypassing

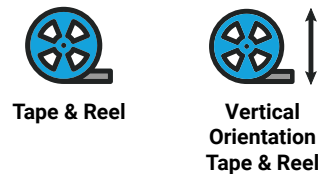
ENVIRONMENTAL TEST

Thermal Shock	MIL-STD-202, Method 107, Condition A
Moisture Resistance	MIL-STD-202, Method 106
Low Voltage Humidity	MIL-STD-202, Method 103, Condition A, with 1.5 Volts DC applied while subjected to an environment of 85°C with 85% relative humidity for 240 hours min.
Life Test	MIL-STD-202, Method 108, for 2000 hours, at 125°C 200% WVDC applied

FEATURES

- Case B Size (.110" x .110")
- Rugged, reliable NP0 dielectric
- Case optimized for highest self resonant frequency
- Capacitance Range 0.1 pF to 1000 pF
- Lowest ESR
- Capable of highest RF Power
- RoHS Compliant/Lead-Free

PACKAGING OPTIONS



ENVIRONMENTAL CHARACTERISTICS

Quality Factor	Greater than 2,000 at 1 MHz (Cap Values $\leq 10\text{pf}$.0010 Max. @ 1MHz Cap Values $> 10\text{pf}$.0005 Max. @ 1MHz)
Temperature Coefficient of Capacitance (TCC)	0 $\pm$ 30 PPM/°C (-55°C to +125°C)
Insulation Resistance (IR)	0.1 pF to 1000 pF: 10 ⁵ Megohms min. @ +25°C at rated WVDC 10 ⁴ Megohms min. @ +125°C at rated WVDC
Working Voltage (WVDC)	See Capacitance Values Table
Dielectric Withstanding Voltage (DWV)	Case B: 250% of rated WVDC for 5 secs
Retrace	Less than $\pm(0.02\%$ or 0.02 pF), whichever is greater
Aging Effects	None
Piezoelectric Effects	None
Capacitance Drift	$\pm(0.02\%$ or 0.02 pF), whichever is greater
Operating Temperature Range	From -55°C to +125°C (No derating of working voltage)
Termination Styles	See Mechanical Configurations
Terminal Strength	Terminations for chips withstand a pull of 5 lbs. min., 15 lbs. typical, for 5 seconds in direction perpendicular to the termination surface of the capacitor. Test per MIL-STD-202, method 211.



The Important Information/Disclaimer is incorporated in the catalog where these specifications came from or available online at www.kyocera-avx.com/disclaimer/ by reference and should be reviewed in full before placing any order.

TDS-RFM-0022 | Rev 4

— RF MICROWAVE PRODUCTS —

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CAPACITANCE VALUES

Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC	Cap. Code	Cap. (pF)	Tol.	Rated WVDC	
0R1	0.1	B	500	2R4	2.4	B, C, D	500	200	20	F, G, J, K, M	500	151	150	F, G, J, K, M	300	
0R2	0.2			2R7	2.7			220	22			161	160			
0R3	0.3	B, C		3R0	3.0			240	24			181	180			
0R4	0.4			3R3	3.3			270	27			201	200			
0R5	0.5	B, C, D		3R6	3.6			300	30			221	220		200	
0R6	0.6			3R9	3.9			330	33			241	240			
0R7	0.7			4R3	4.3			360	36			271	270			
0R8	0.8			4R7	4.7			390	39			301	300			
0R9	0.9			5R1	5.1			430	43			331	330			
1R0	1.0			5R6	5.6			470	47			361	360			
1R1	1.1			6R2	6.2			510	51			391	390			
1R2	1.2			6R8	6.8	560		56	431			430				
1R3	1.3	B, C, D		7R5	7.5	620		62	471			470	100			
1R4	1.4			8R2	8.2	680		68	511			510				
1R5	1.5			9R1	9.1	750		75	561			560				
1R6	1.6			100	10	820		82	621			620	50			
1R7	1.7			110	11	910		91	681			680				
1R8	1.8			120	12	101		100	751			750				
1R9	1.9			130	13	111		110	821			820				
2R0	2.0			150	15	121		120	911		910					
2R1	2.1			160	16	131		130	102		1000					
2R2	2.2															

VRMS = 0.707 X WVDC

• SPECIAL VALUES, TOLERANCES AND MATCHING AVAILABLE. PLEASE CONSULT FACTORY.

HOW TO ORDER

Series 800

Case Size B

See mechanical dimensions below

Capacitance 910

EIA Capacitance Code in pF.
First two digits = significant figures or "R" for decimal place.
Third digit = number of zeros or after "R" significant figures

Capacitance Tolerance Code J

Code	B	C	D	F	G	J	K	M
Tol.	±0.1 pF	±0.25 pF	±0.5 pF	±1%	±2%	±5%	±10%	±20%

Rated WVDC 500

Termination Code T

Packaging T

T = Tape and Reel, 500 pc qty
TV = Vertical Tape and Reel, 500 pc qty
Please see last Column Mechanical Configuration Table for other options

Laser Marking (Optional)

WVDC

Please see 2nd Column Mechanical Configuration Table

The above part number refers to a 800 B Series (case size B) 91 pF capacitor, J tolerance (±5%), 500 WVDC, with T termination (Tin Plated over Nickel Barrier Termination, RoHS Compliant), laser marking and tape and reel packaging.

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MECHANICAL CONFIGURATION

Series & Case Size	Term. Code	Case Size & Type	Outline ES W/T is a Termination Surface	Body Dimensions inches (mm)			Lead and Termination Dimensions and Material		Pkg Type & Qty	Pkg Code
				Length (L)	Width (W)	Thickness (T)	Overlap (Y)	Materials		
800B	T	B Solderable Nickel Barrier		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)	.070 (1.78) max.	.015 (0.38) ±.010 (0.25)	RoHS Compliant Tin Plated over Nickel Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV
800B	W	B Solder Plate		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)			Tin/ Lead, Solder Plated over Nickel Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV

NON-MAGNETIC CONFIGURATION

Series & Case Size	Term. Code	Case Size & Type	Non-Magnetic Configuration	Body Dimensions inches (mm)			Lead and Termination Dimensions and Material		Pkg Type & Qty	Pkg Code
				Length (L)	Width (W)	Thickness (T)	Overlap (Y)	Materials		
800B	TN	B Non-Mag Solderable Barrier		.110+.020 -.010 (2.79+0.51-0.25)	.110 ±.015 (2.79 ±0.38)	.070 (1.78) max.	.015 (0.38) ±.010 (0.25)	RoHS Compliant Tin Plated over Non-Magnetic Barrier Termination	T&R, 1000 or 500 pcs Vertical T&R, 1000 or 500 pcs	T1K or T TV1K or TV

SUGGESTED MOUNTING PAD DIMENSIONS

Horizontal Electrode Orientation Vertical Electrode Orientation

Case B Vertical Mount					
Cap Value .43	Pad Size	A Min.	B Min.	C Min.	D Min.
All Values	Normal	.090 (2.29)	.050 (1.27)	.075 (1.91)	.175 (4.45)
	High Density	.070 (1.78)	.030 (.762)	.075 (1.91)	.135 (3.43)

inches (mm)

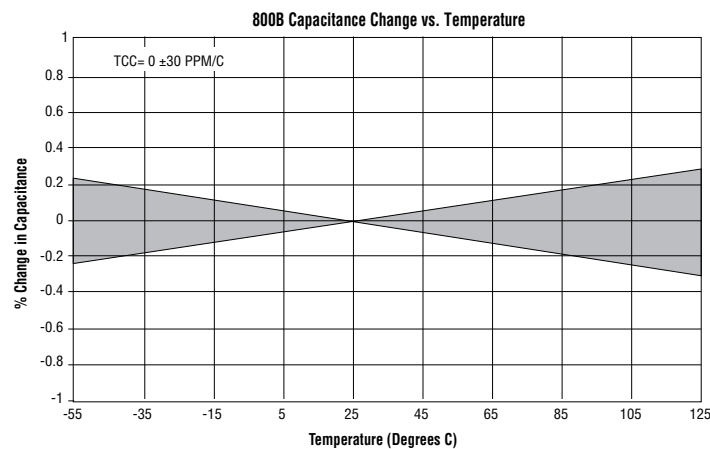
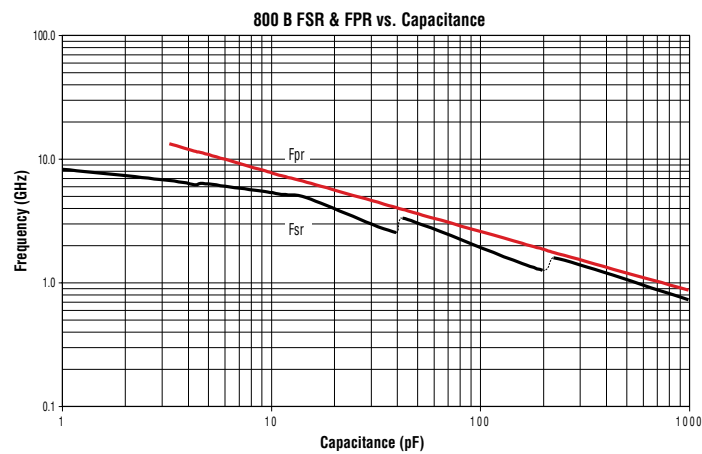
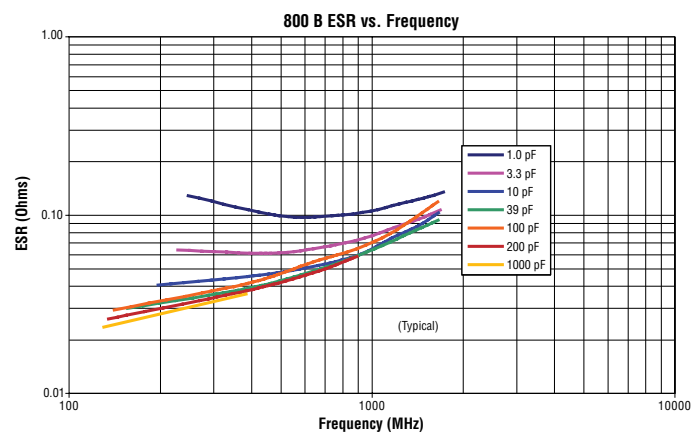
Case B Horizontal Mount					
Cap Value .43	Pad Size	A Min.	B Min.	C Min.	D Min.
All Values	Normal	.130 (3.30)	.050 (1.27)	.075 (1.91)	.175 (4.45)
	High Density	.110 (2.79)	.030 (.762)	.075 (1.91)	.135 (3.43)

inches (mm)

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PERFORMANCE DATA



RF/Microwave Capacitors

RF/Microwave Multilayer Capacitors (MLC)

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SAMPLE KITS

Kit #	RoHS Compliant	Item Number	Description	Cap. Value Range (pF)	Cap Value (pF)	Tol.	Price
Kit 83T		DK0083T	800B NP0 Ceramic High RF Power MLCs 16 different values, 15 pcs. min. per value	1.0 to 10	1.0, 1.2, 1.5, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3..... 3.9, 4.7, 5.6, 6.8, 8.2 10	±0.1 ±0.25 ±5%	\$165.00
Kit 84T		DK0084T	800B NP0 Ceramic High RF Power MLCs 16 different values, 15 pcs. min. per value	10 to 100	10, 12, 15, 18, 20, 22, 24, 27, 30, 33, 39, 47, 56, 68, 82, 100	±0.1	\$165.00
Kit 85T		DK0085T	800B NP0 Ceramic High RF Power MLCs 16 different values, 15 pcs. min. per value	100 to 1000	100, 120, 150, 180, 200, 220, 240, 270, 300, 330, 390, 470..... 560, 680, 820, 1000.....	±5% ±10%	\$165.00